Engineering Chemistry(CHY1001)

B.Tech. (Chemistry Cycle)

Module-4

Faculty of Science, Technology and Architecture

School of Physical and Biological Sciences

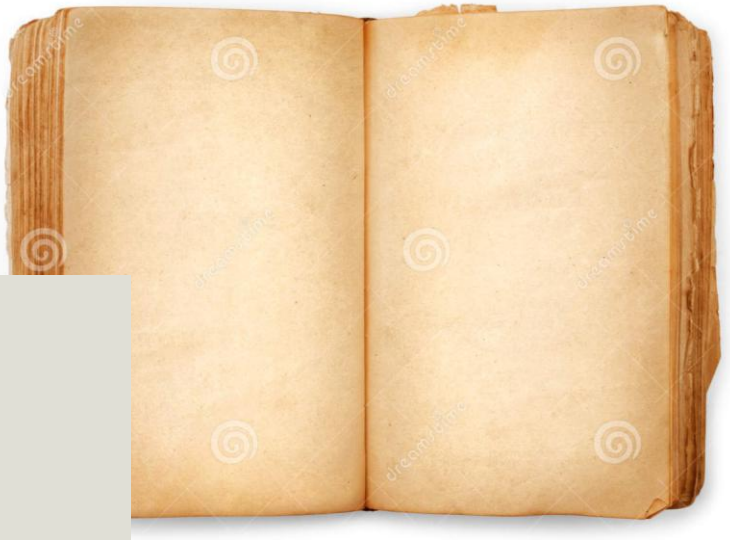
Department of Chemistry

Manipal University Jaipur



MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)





What is Corrosion

Corrosion (n)

- The chemical or electrochemical reaction between a material and its environments that produces a deterioration of the material and its properties.











Corrosion

Corrosion can be considered as the **degradation of a material by chemical means.**

- Although all materials will degrade in some way, the term corrosion is usually applied to metals, hence metallic corrosion.
- When considering metals, corrosion is essentially the reverse of the refining process. When metals are extracted from the earth they are refined into pure metals from the various ores **(metal oxides, hydroxides, carbonates, etc**).
- As a result of corrosion, the metal reacts with chemicals in the environment and progressively returns to the combined form, rust is a form of iron oxide.



(Reverse process of metal extraction)

1. **Acid theory:**

- The presence of acid is essential.
- Example: rusting of iron.

$$Fe \xrightarrow{O_2, CO_2, H_2O} Fe(HCO_3)_2 \xrightarrow{\text{Oxidation}} Fe(OH)CO_3 \rightarrow 2\ Fe(OH)_3$$

2. **Direct chemical attack:** This theory explains the so-called chemical or dry corrosion. Example: direct chemical attack by dry gases

3. **The electrochemical theory:** This theory explains the indirect or wet corrosion. The modern electrochemical theory is based on Nernst theory.

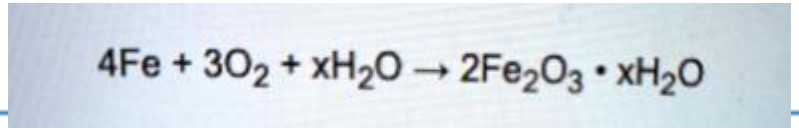
Acid theory:

- This theory suggests that the presence of acid (such as carbonic acid) is essential for corrosion.
- This theory is particularly applicable to rusting of iron in the atmosphere.
- The rusting is due to continued action of oxygen, carbon dioxide and moisture, converting the metal into soluble ferrous bicarbonate which is further oxidized to basic ferric carbonate and finally to hydrated ferric oxide.

$$Fe + \frac{1}{2}O_2 + 2\,CO_2 + H_2O \rightarrow Fe(HCO_3)_2$$

$$Fe(HCO_3)_2 + O_2 \rightarrow Fe(OH)CO_3 + CO_2 + H_2O$$

$$2\,Fe(OH)CO_3 + 2\,H_2O \rightarrow 2\,Fe(OH)_3 + 2CO_2$$



Theories of corrosion

DRY CORROSION(OR)CHEMICAL CORROSION

1. OXIDATION CORROSION
2. CORROSION BY OTHER GASES
3. LIQUID-METAL CORROSION

WET CORROSION (OR) ELECTROCHEMICAL CORROSION

1. ELECTROCHEMICAL CORROSION BY EVOLUTION OF HYDROGEN AND ABSORPTION OF OXYGEN.
2. GALVANIC CORROSION
3. CONCENTRATION CELL CORROSION

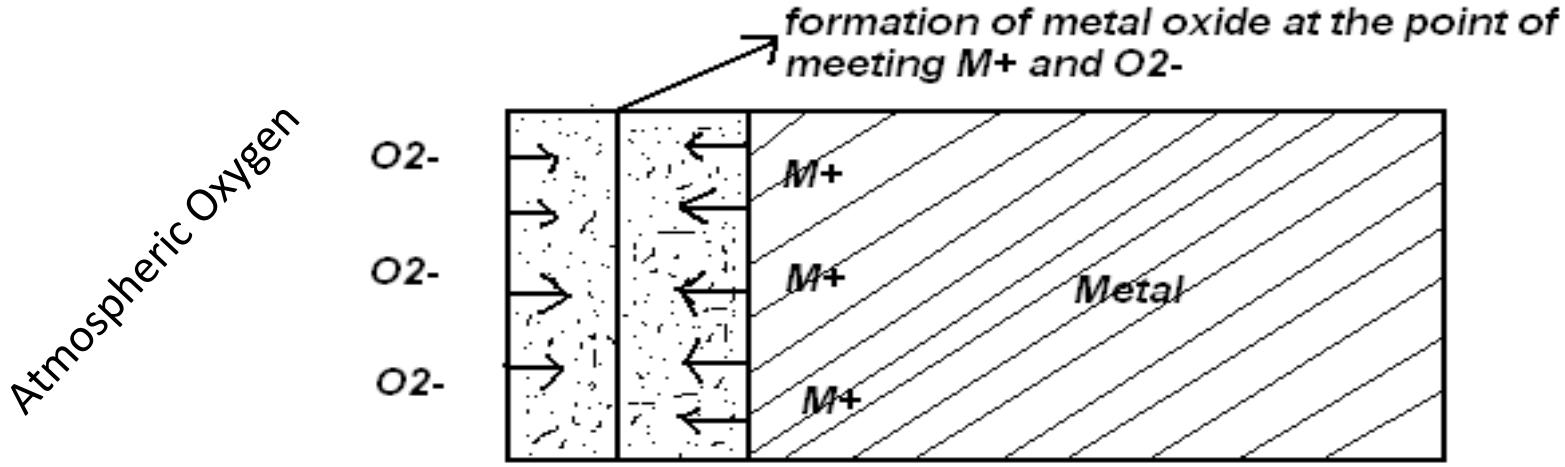
Chemical (or) Dry Corrosion

What is it?

The direct chemical attack on the atmospheric content on the metal surfaces in the absence of moisture

1. **Oxidation Corrosion:** Direct action of oxygen at high (or) low temperature on metal surfaces.
2. **Corrosion by other Gases:** Attack of gases like SO_2, CO_2, H_2S, Cl_2, F_2 etc. on metal surfaces.
3. **Liquid Metal Corrosion :** Attack of inorganic liquid metals on solid metallic surfaces.

1. Oxidation Corrosion



Oxidation mechanism of metals

$$2\,M \rightarrow 2\,M^{2+} + 4\,e^{-}$$ → (Loss of electrons = **Oxidation**)

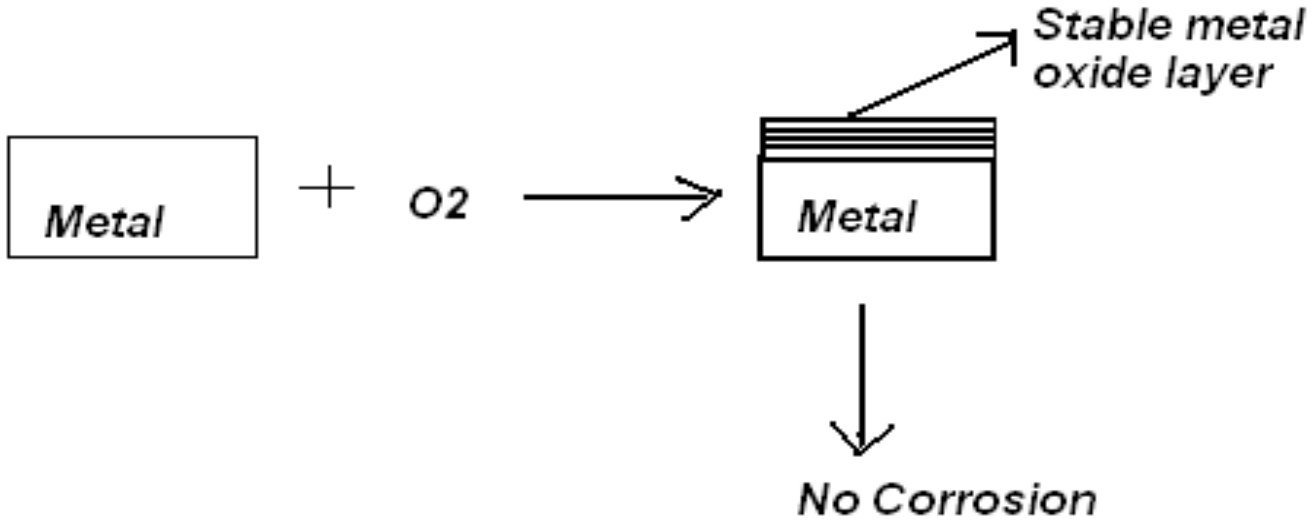
$$O_2 + 4\,e^{-} \rightarrow 2\,O^{2-}$$ → (Gain of electrons = **Reduction**)

$$2\,M + O_2 \rightarrow 2\,M^{2+} + 2\,O^{2-}$$

1. Oxidation Corrosion

If the formed metal oxide is stable,

further corrosion of metal is prevented



Stable metal oxide layer

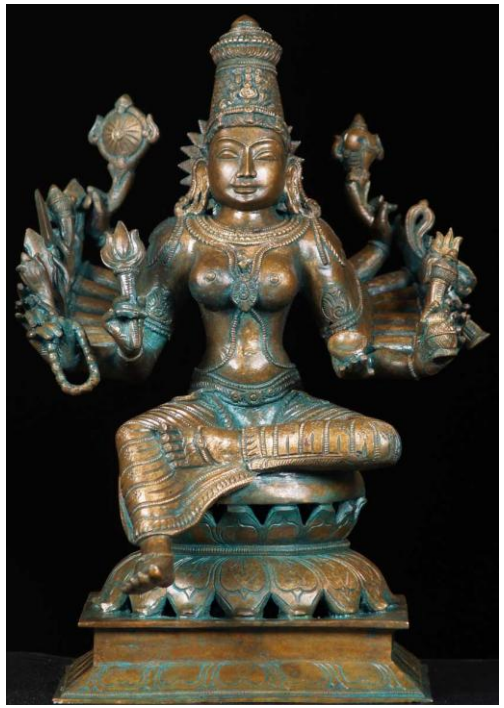
(Example: Al, Sn, Pb, Cu)

1. Oxidation Corrosion

If the formed metal oxide is stable, further corrosion of metal is prevented



(Example: Copper)



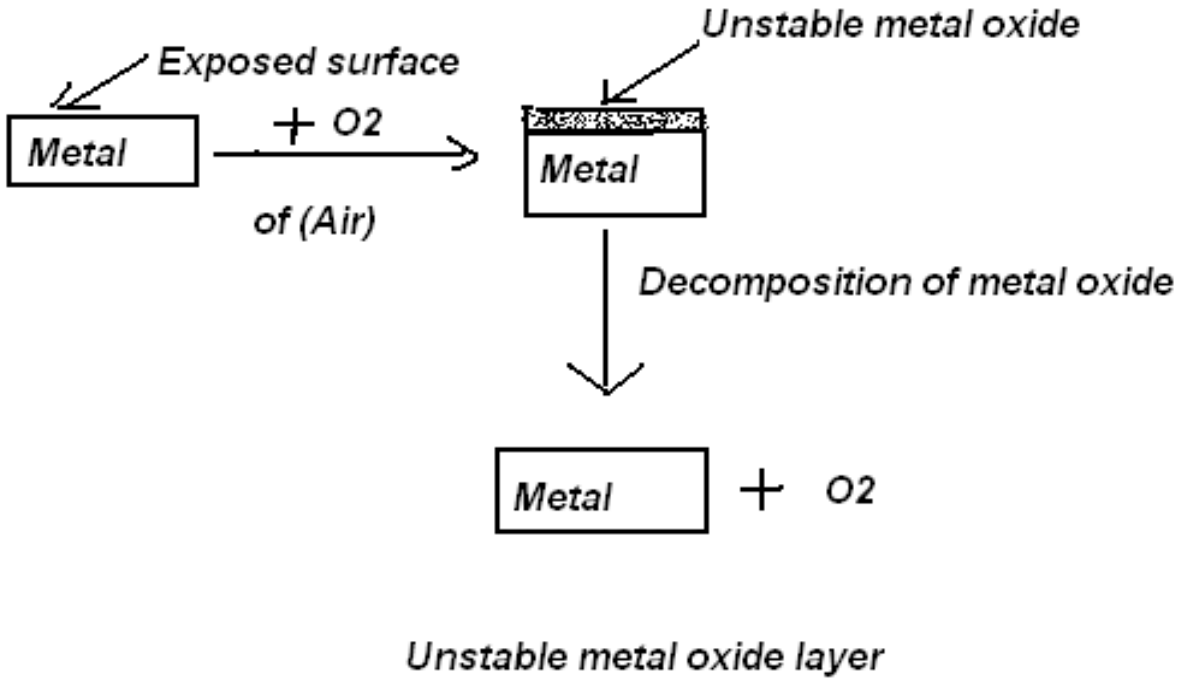


(Example: Bronze)

1. Oxidation Corrosion

If the formed metal oxide is unstable,

corrosion does not occur*

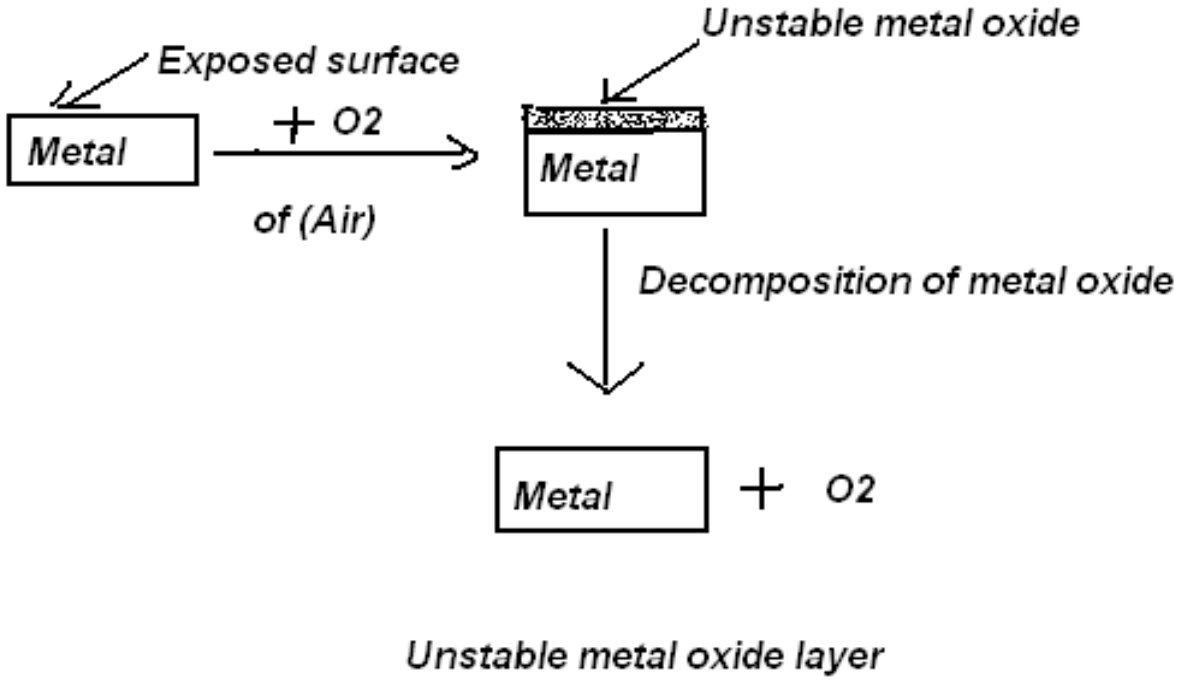


Unstable metal oxide layer

1. Oxidation Corrosion

If the formed metal oxide is unstable,

corrosion does not occur*



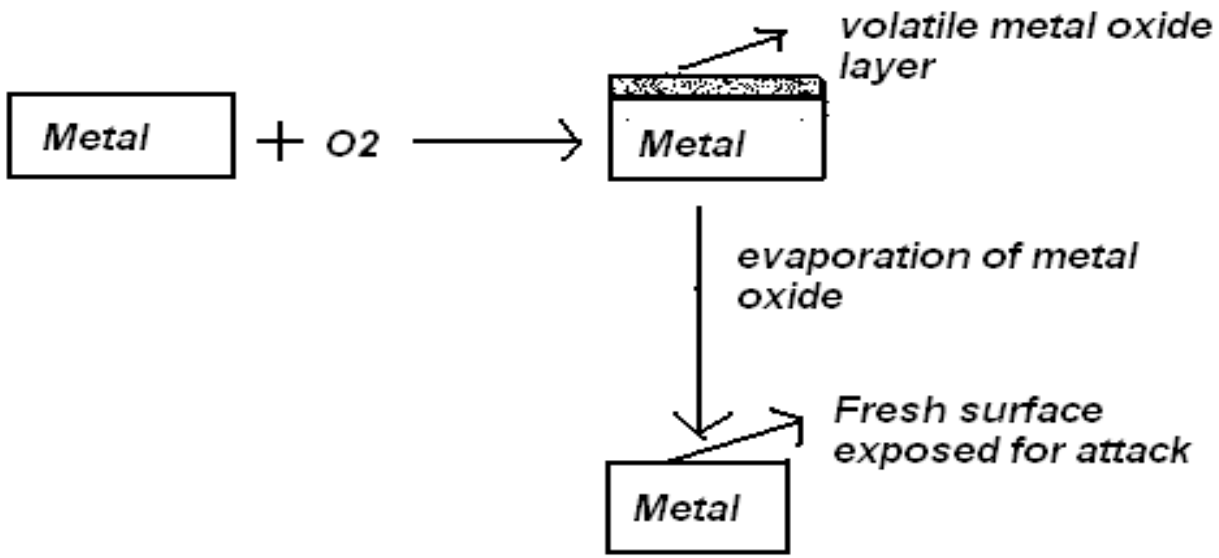
Unstable metal oxide layer

(Example: Ag, Au, Pt) → Nobel metals

1. Oxidation Corrosion

If the formed metal oxide is volatile,

surface is rapidly exposed, and evaporated

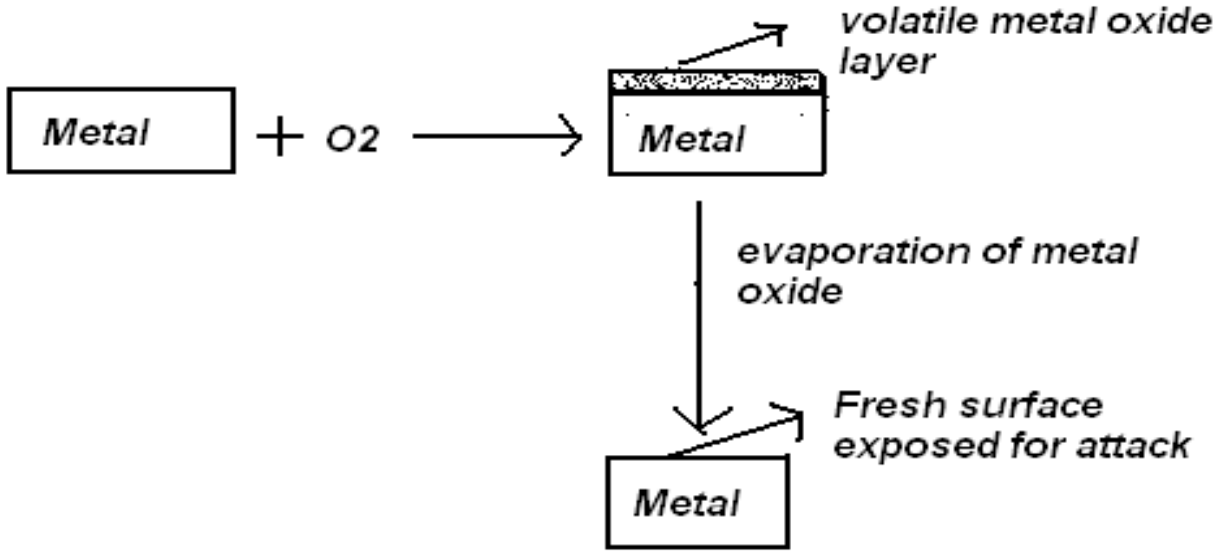


volatile metal oxide layer

1. Oxidation Corrosion

If the formed metal oxide is volatile,

surface is rapidly exposed, and evaporated



volatile metal oxide layer

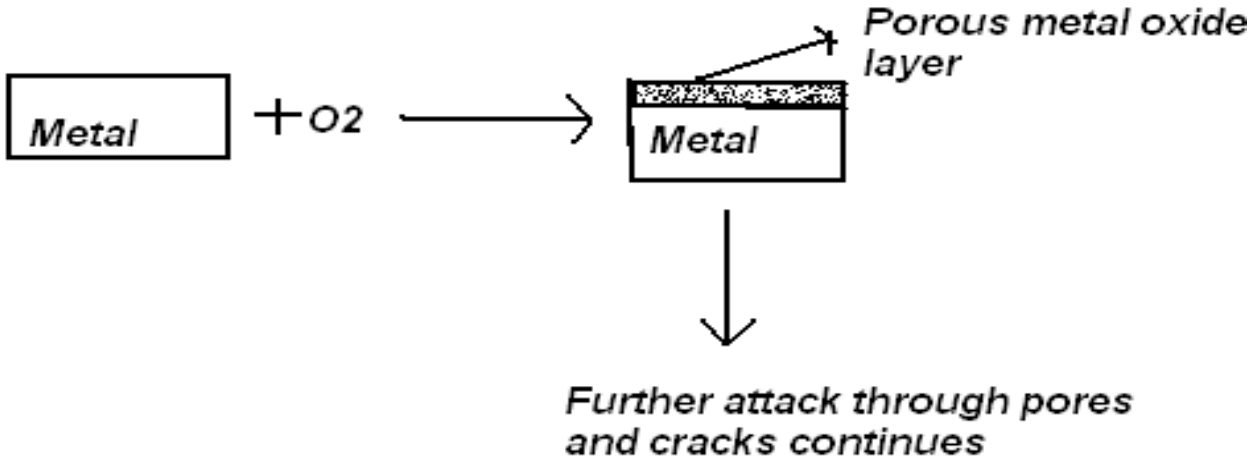


(Example: Mo)

1. Oxidation Corrosion

If the formed metal oxide is porous,

surface is rapidly exposed, and evaporated

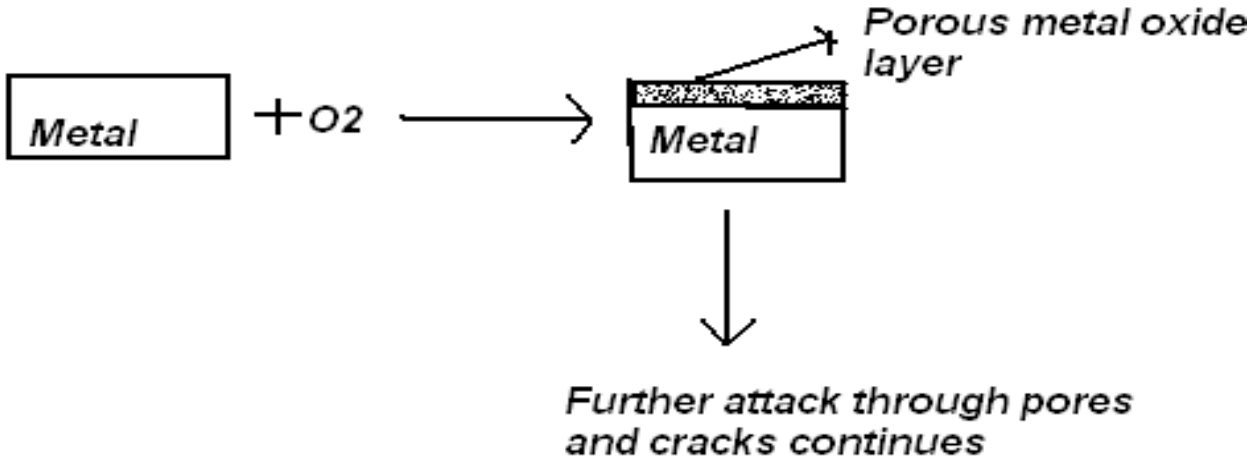


Porous metal oxid layer

1. Oxidation Corrosion

If the formed metal oxide is porous,

surface is rapidly exposed, and evaporated



Porous metal oxid layer



(Example: Iron, Steel)

Oxidative Corrosion

Four types of oxides:

If the formed metal oxide is stable*,
further corrosion of metal is prevented
Example: Cu, Al, Pb, Sn

If the formed metal oxide is unstable,
corrosion does not occur*
Example: Au, Ag, Pt

If the formed metal oxide is volatile,
surface is rapidly exposed, and evaporated
Example: Mo

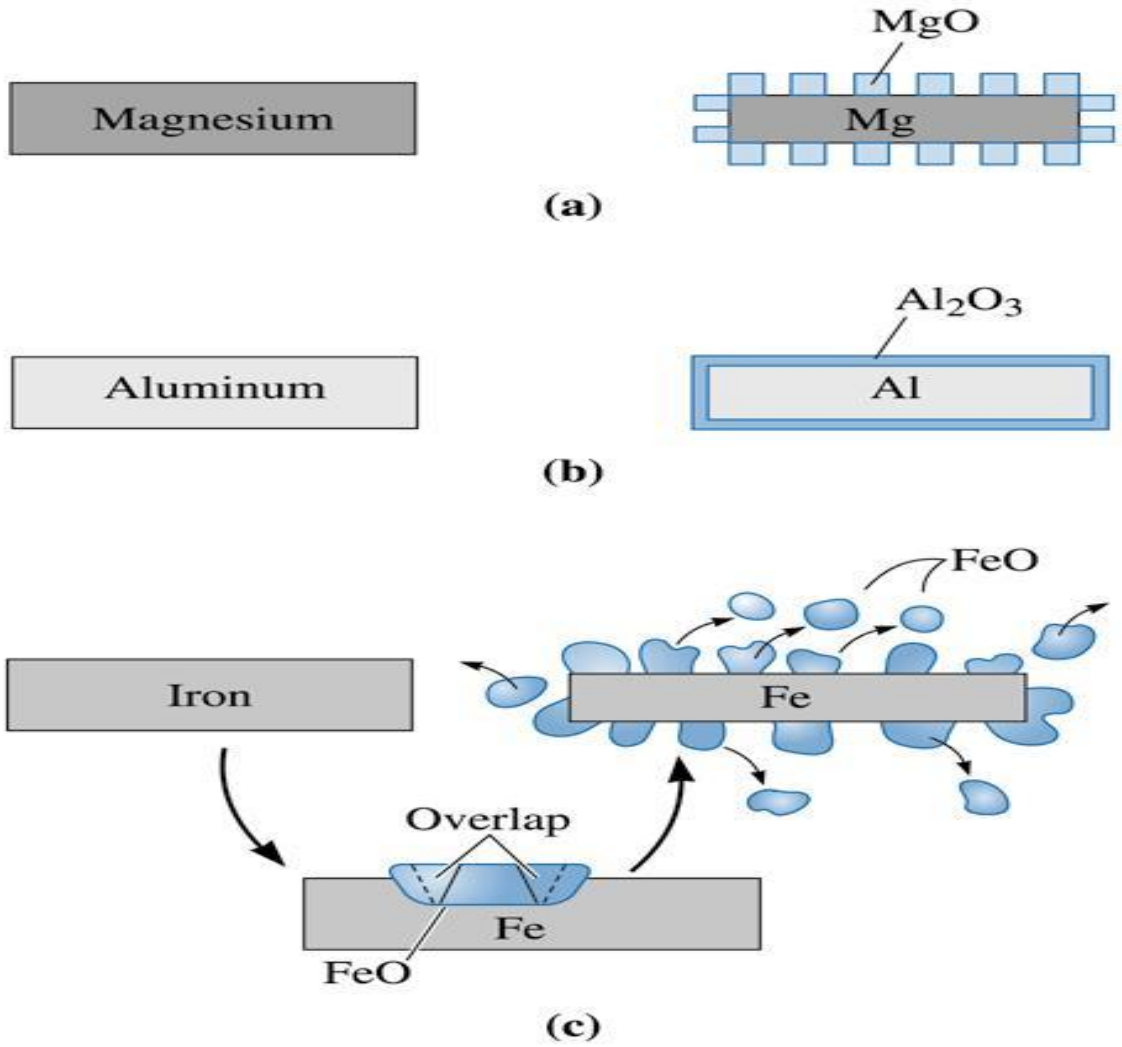
If the formed metal oxide is porous,
surface is rapidly exposed, and evaporated
Example: Fe

Three types of oxides may form, depending on the volume ratio between the metal and the oxide:

a. Magnesium produces a porous oxide film,

b. Aluminum forms a protective, adherent, nonporous oxide film,

c. Iron forms an oxide film that spills off the surface and provides poor protection.



Pilling-Bedworth Ratio

The **Pilling-Bedworth ratio (P-B ratio)**, in corrosion of metals, is the ratio of the volume of the elementary cell of a metal oxide to the volume of the elementary cell of the corresponding metal (from which the oxide is created).

$$\mathbf{R_{PB}} = \frac{V_{Oxide}}{V_{Metal}} = \frac{\frac{M_{Oxide}}{\rho_{Oxide}}}{n \times \frac{M_{Metal}}{\rho_{Metal}}} = \frac{M_{Oxide} \times \rho_{Metal}}{n \times M_{Metal} \times \rho_{Oxide}}$$

where,

R_{PB} : is the Pilling-Bedworth ratio,
V : the molar volume
ρ : density,
M : the atomic or molecular mass,
n : number of atoms of metal per one molecule of the oxide

Pilling-Bedworth Ratio

On the basis of measurements, the following connection can be shown:

$$\mathbf{R_{PB}} = \frac{V_{Oxide}}{V_{Metal}} = \frac{M_{Oxide} \times \rho_{Metal}}{n \times M_{Metal} \times \rho_{Oxide}}$$

$\mathbf{R_{PB}} < \mathbf{1}$: the oxide coating layer is too thin, likely broken and provides no protective effect (for example magnesium)

$\mathbf{R_{PB}}$: between 1-2: the oxide coating provides protective effect

$\mathbf{R_{PB}} > \mathbf{2}$: Non protective

However, the exceptions to the above P-B ratio rules are numerous. Many of the exceptions can be attributed to the mechanism of the oxide growth: the underlying assumption in the P-B ratio is that oxygen needs to diffuse through the oxide layer to the metal surface; in reality, it is often the metal ion that diffuses to the air-oxide interface.

Calculation of Pilling-Bedworth Ratio

$$R_{PB} = \frac{V_{Oxide}}{V_{Metal}} = \frac{M_{Oxide} \times \rho_{Metal}}{n \times M_{Metal} \times \rho_{Oxide}}$$

Oxide	Oxide Density (g/cc)	Atomic weight of Metal	Metal Density (g/cc)	Molecular Weight of Oxide	P-B Ratio
MgO	3.65	24.3	1.74	40.3	0.79
CaO	3.40	40.08	1.55	56.1	0.63
Al_2O_3	3.70	27.00	2.70	102.0	1.37

Pilling-Bedworth Ratio

$$\mathbf{R_{PB}} = \frac{V_{Oxide}}{V_{Metal}} = \frac{\frac{M_{Oxide}}{\rho_{Oxide}}}{n \times \frac{M_{Metal}}{\rho_{Metal}}} = \frac{M_{Oxide} \times \rho_{Metal}}{n \times M_{Metal} \times \rho_{Oxide}}$$

where,

R_{PB} : is the Pilling-Bedworth ratio, V : the molar volume, ρ : density, M : the atomic or molecular mass, n : number of atoms of metal per one molecule of the oxide

Oxide	Oxide Density (g/cc)	Atomic Weight of Metal	Metal Density (g/cc)	Molecular Weight of Oxide	P-B Ratio
Cr_2O_3	5.22	52	7.19	152.0	2.01
Na_2O	2.27	23	0.97	62.0	0.57
NiO	6.67	58.69	8.91	74.7	1.70
K_2O	2.13	39.09	0.86	94.2	0.48

2. Corrosion by gases

The gases such as SO_2, CO_2, H_2S, Cl_2, F_2 etc., when come in direct contact with metal surface corrosion occurs.

- The extent of corrosion depends on the chemical affinity between the metal and the gas concerned.
- The prevention of metal corrosion can be known from the nature of corrosion product.

Whether the layer of corrosion product is protecting or non-protecting in nature.

- If the formed corrosion product is protecting (or) non porous metal is prevented. Ex: AgCl layer on metallic silver by the action of Cl_2 gas.
- If the formed corrosion product is non protecting (or) porous , the corrosion of metals occurs non stop. Ex: H_2S gas attacks on steel at high temperature forming FeS , a corrosion product which is porous.

3. Liquid metal corrosion

The chemical action of the flowing liquid metal at high temperature, on a solid metal or alloy produces liquid metal corrosion.

There are two reasons for this corrosion –

- Dissolution of the solid metal by liquid metal

- Internal penetration of the liquid metal into the solid phase, weakening the solid metal.

Example: liquid Na used as a coolant in a nuclear plants, these causes cadmium corrosion.

Mechanism of wet corrosion

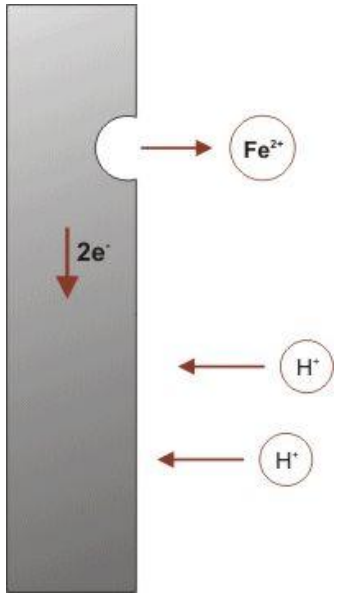
1. The formation of anodic and cathodic areas or parts in contact with each other.
2. Presence of a conducting medium.
3. Corrosions of anodic areas only.
4. Formation of corrosion product somewhere between anodic and cathodic areas.

1) Mechanism of wet or electrochemical corrosion

The **metal surface** undergoes an *electrochemical reaction* with the **moisture and oxygen** in the atmosphere. This theory is known as electrochemical theory of corrosion.

Mechanism :

- **Anodic reaction :** Oxidation of metal (corrosion)
- **Cathodic reaction :** Consumption of electrons



Corrosion always occurs at the anode

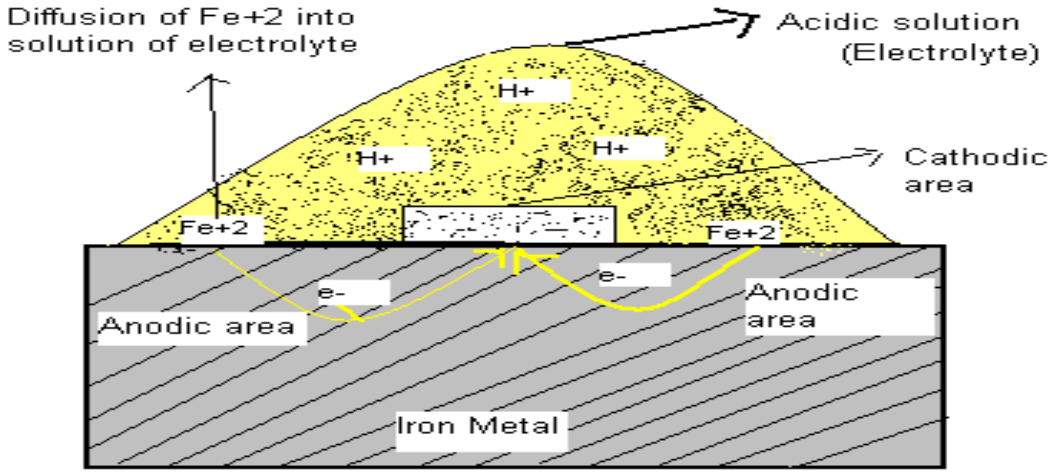
1) Mechanism of wet or electrochemical corrosion

Wet corrosion takes place by the following two ways based on the nature of medium:

1) Anodic reaction: $Fe\ (s) \rightarrow Fe^{2+}\ (aq) + 2e^-$

2) Cathodic reaction: a) Evolution of H_2 (acidic medium) in absence of oxygen

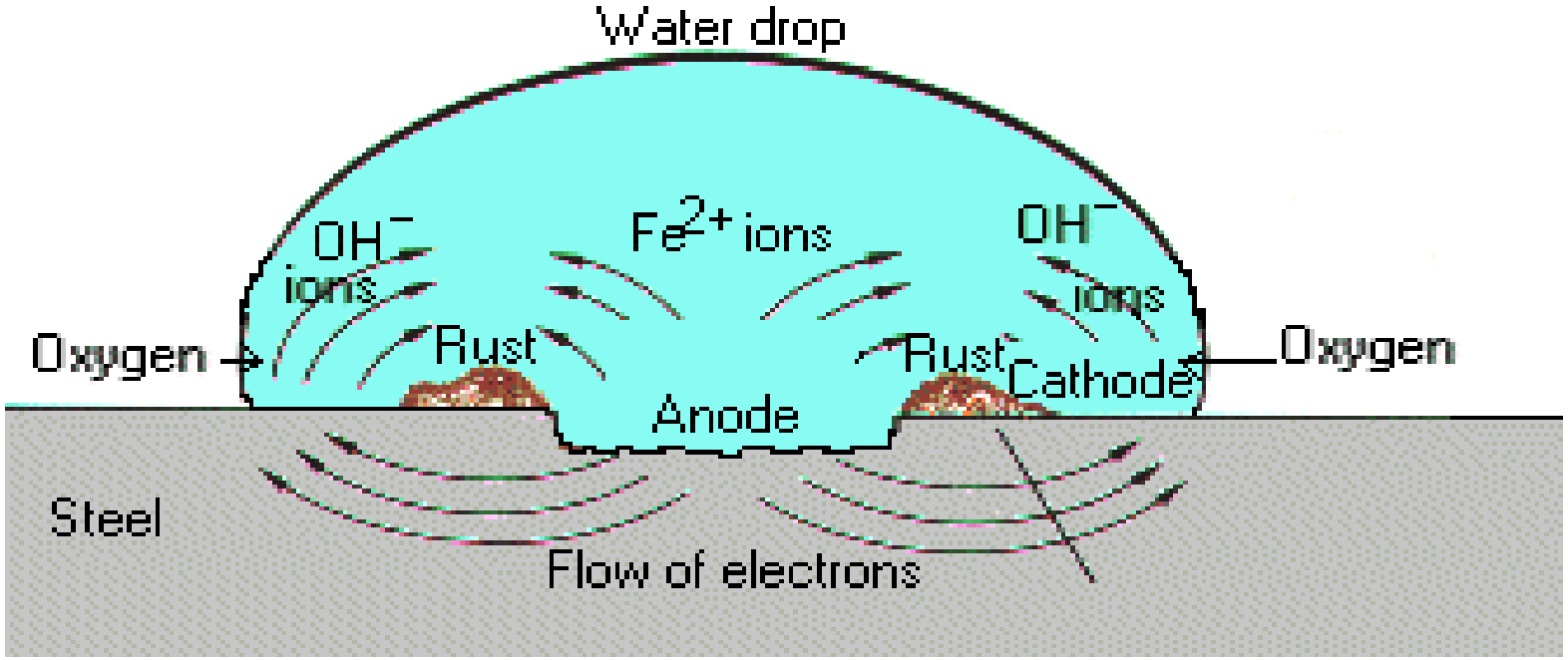
$$2\ H^+\ (aq) + 2e^- \rightarrow H_2$$



Displacement of hydrogen ions from the acidic solution by metal ions)

b) Absorption of O_2 : Neutral medium

Cathode: ½ $O_2(g)$ + H_2O (aq) +2 e- → $2OH^-$

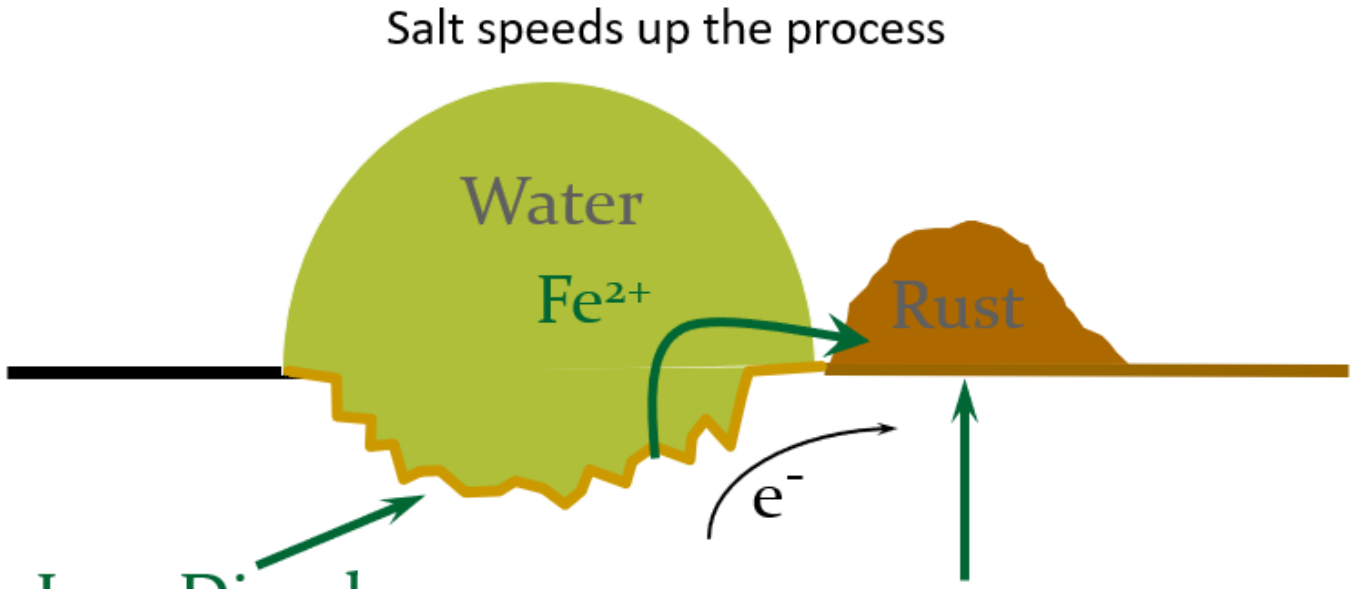


If the solution is aerated and almost neutral, O_2 is reduced in presence of water to OH^-

c. If the solution is deaerated and almost neutral, H_2 is liberated along with OH^-

Cathode: $H_2O\ (aq) + 2\ e^- \rightarrow H_2 + 2OH^-$

Final Step

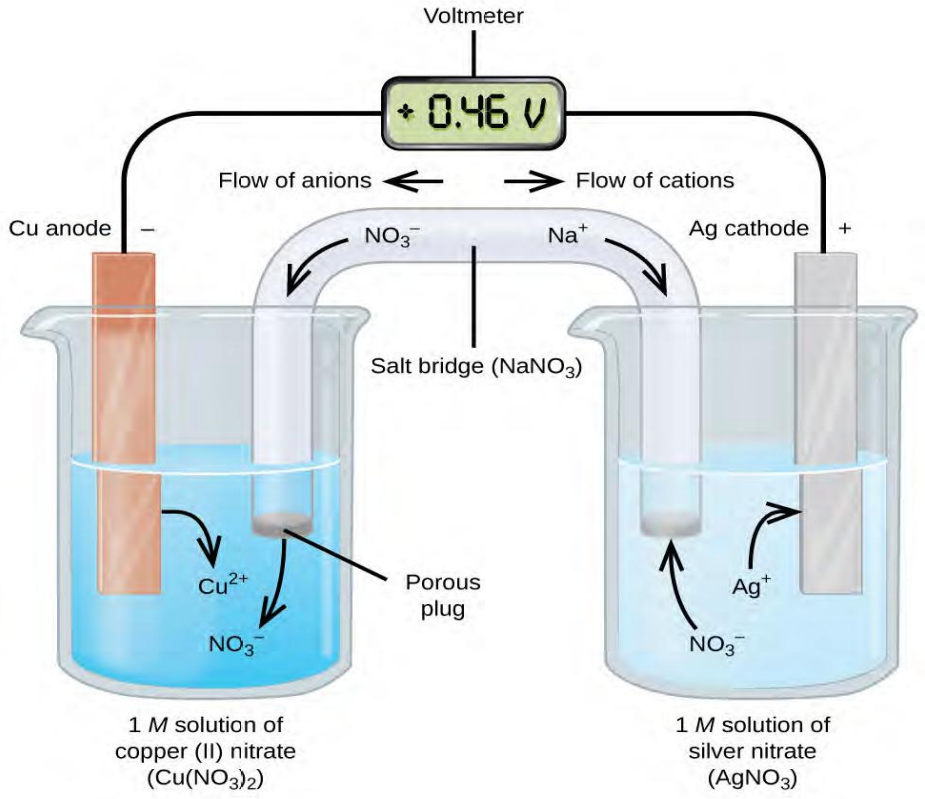


$$Fe^{2+} + 4\ OH^- \longrightarrow 2Fe(OH)_2$$

$$2Fe(OH)_2 + \tfrac{1}{2}\ O_2 + (x-2)\ H_2O \longrightarrow Fe_2O_3.\ xH_2O$$

(2) Galvanic Corrosion

Galvanic Corrosion is an electrochemical process in which one metal corrodes preferentially to another when both metals are in electrical contact and immersed in an electrolyte.

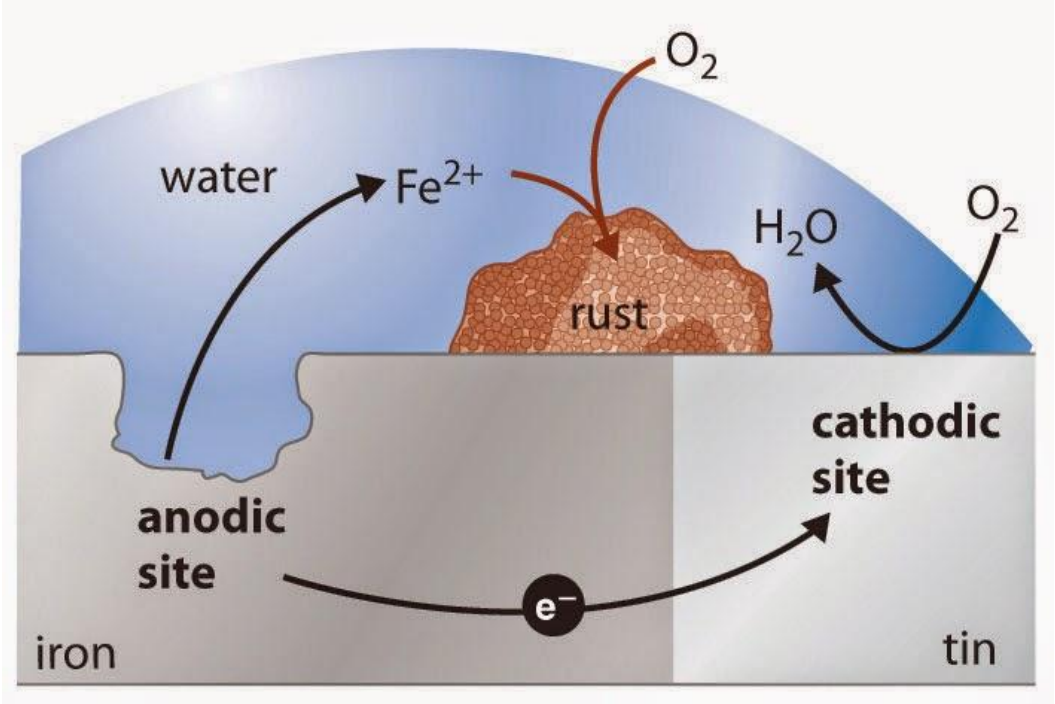


Galvanic cell

Galvanic Corrosion

Galvanic corrosion is an electrochemical process in which one metal corrodes preferentially to another when both metals are in electrical contact and immersed in an electrolyte.





Galvanic Corrosion

- Possibility when two dissimilar metals are electrically connected in an electrolyte.
- Results from a difference in oxidation potentials of metallic ions between two or more metals. The greater the difference in oxidation potential, the greater the galvanic corrosion.
- Galvanic coupling is a galvanic cell in which the anode is the less corrosion resistant metal than the cathode
- The less noble metal will corrode (i.e. will act as the anode) and the more noble metal will not corrode (acts as cathode).
- Perhaps the best known of all corrosion types is galvanic corrosion, which occurs at the contact point of two metals or alloys with different electrode potentials.

Standard electrode potential

TABLE 18.1 Standard Reduction Potentials at 25°C

	Reduction Half-Reaction		E° (V)	
Stronger oxidizing agent	$F_2(g) + 2\,e^-$	$\longrightarrow 2\,F\,(aq)$	2.87	Weaker reducing agent
	$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^-$	$\longrightarrow 2\,H_2O(l)$	1.78	
	$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^-$	$\longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$	1.51	
	$Cl_2(g) + 2\,e^-$	$\longrightarrow 2\,Cl^-(aq)$	1.36	
	$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^-$	$\longrightarrow 2\,Cr^{3+}(aq) + 7\,H_2O(l)$	1.33	
	$O_2(g) + 4\,H^+(aq) + 4\,e^-$	$\longrightarrow 2\,H_2O(l)$	1.23	
	$Br_2(l) + 2\,e^-$	$\longrightarrow 2\,Br^-(aq)$	1.09	
	$Ag^+(aq) + e^-$	$\longrightarrow Ag(s)$	0.80	
	$Fe^{3+}(aq) + e^-$	$\longrightarrow Fe^{2+}(aq)$	0.77	
	$O_2(g) + 2\,H^+(aq) + 2\,e^-$	$\longrightarrow H_2O_2(aq)$	0.70	
	$I_2(s) + 2\,e^-$	$\longrightarrow 2\,I^-(aq)$	0.54	
	$O_2(g) + 2\,H_2O(l) + 4\,e^-$	$\longrightarrow 4\,OH^-(aq)$	0.40	
	$Cu^{2+}(aq) + 2\,e^-$	$\longrightarrow Cu(s)$	0.34	
	$Sn^{4+}(aq) + 2\,e^-$	$\longrightarrow Sn^{2+}(aq)$	0.15	
	$2\,H^+(aq) + 2\,e^-$	$\longrightarrow H_2(g)$	0	
	$Pb^{2+}(aq) + 2e^-$	$\longrightarrow Pb(s)$	−0.13	
	$Ni^{2+}(aq) + 2\,e^-$	$\longrightarrow Ni(s)$	−0.26	
	$Cd^{2+}(aq) + 2\,e^-$	$\longrightarrow Cd(s)$	−0.40	
	$Fe^{2+}(aq) + 2\,e^-$	$\longrightarrow Fe(s)$	−0.45	
	$Zn^{2+}(aq) + 2\,e^-$	$\longrightarrow Zn(s)$	−0.76	
	$2\,H_2O(l) + 2\,e^-$	$\longrightarrow H_2(g) + 2\,OH^-(aq)$	−0.83	
	$Al^{3+}(aq) + 3\,e^-$	$\longrightarrow Al(s)$	−1.66	
	$Mg^{2+}(aq) + 2\,e^-$	$\longrightarrow Mg(s)$	−2.37	
	$Na^+(aq) + e^-$	$\longrightarrow Na(s)$	−2.71	
Weaker oxidizing agent	$Li^+(aq) + e^-$	$\longrightarrow Li(s)$	−3.04	Stronger reducing agent

The Galvanic series

The galvanic series (or electro-potential series) determines the **nobility** of metals and semi-metals.

When two metals are submerged in an electrolyte, while electrically connected, the less noble (base) will experience galvanic corrosion.

The rate of corrosion is determined by

- the electrolyte and
- the difference in nobility.

The difference can be measured as a difference in voltage potential.

Galvanic reaction is the principle upon which batteries are based.

Design for Galvanic Corrosion

Severity of Galvanic corrosion depends on:

- Amount of separation in the galvanic series
- Relative surface areas of the two. Severe corrosion if anode area (area eaten away) is smaller than the cathode area.

Big Cathode, Small Anode = Big Trouble

It does **NOT** depend upon:

- Amount of contact
- Volume
- Mass

Example



Iron screws and brass



Iron screw in Mg

Dissimilar metals, the damage occurs at the anode.

Mercury
Platinum
Gold
Zirconium Graphite
Titanium
Hastelloy C Monel
Stainless Steel (316-passive)
Stainless Steel (304-passive)
Stainless Steel (400-passive)
Nickel (passive oxide)
Silver
Hastelloy 62Ni, 17Cr
Silver solder
Inconel 61Ni, 17Cr
Aluminum (passive Al_2O_3)
70/30 copper-nickel
90/10 copper-nickel
Bronze (copper/tin)
Copper
Brass (copper/zinc)
Alum Bronze Admiralty Brass
Nickel
Naval Brass Tin
Lead-tin
Lead
Hastelloy A
Stainless Steel (active)
316 404 430 410
Lead Tin Solder
Cast iron
Low-carbon steel (mild steel)
Manganese Uranium
Aluminum Alloys
Cadmium
Aluminum Zinc
Beryllium
Magnesium

PASSIVE – will not corrode – act as cathode. These elements are least likely to give up electrons!

ACTIVE – will corrode – act as anode. These elements most likely to give up electrons!

Prevention:

- Material Selection: Do not connect dissimilar metals! Or if you can't avoid it:
  - Try to electrically isolate one from the other (rubber gasket).
  - **Make the anode large and the cathode small**
    - Bad situation: Steel siding with aluminum fasteners
    - Better: Aluminum siding with steel fasteners
- Eliminate electrolyte
- Galvanic of anodic protection

Question

An iron rod and a copper rod are immersed in a conducting solution and two of the electrodes are connected by a wire. The surface area of iron rod and copper rod are x cm^2 and y cm^2 respectively. Consider the two cases-

(i) x: y > 1 and

(ii) x: y < 1

- Which one of the metals will be corroded?
- Among the two cases where the corrosion will be more and why?

(3) Concentration cell corrosion

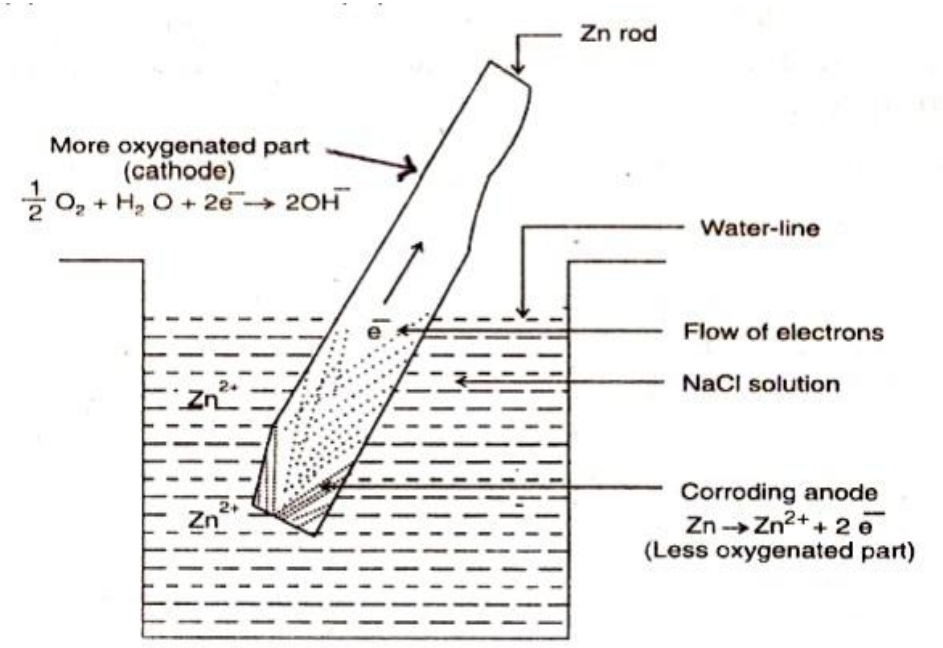
Because of differential aeration, the concentration of O_2 is varied and causes Oxidation of metal .

OXIDATION: $Zn\ (s) \rightarrow Zn^{2+} + 2e^-$

REDUCTION: $\frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2OH^-$

$$Zn^{2+} + 2OH^- \rightarrow Zn(OH)2$$

(Corrosion product)



Examples:

1. Pitting corrosion
2. Water line corrosion
3. Crevice corrosion

3.a) Water line corrosion





When water is stored in a metallic tank, it is observed that the metal below the waterline gets corroded.

- The metal below the waterline is poorly oxygenated and acts as an anode.
- The metal above the waterline is highly oxygenated and acts as a cathode.

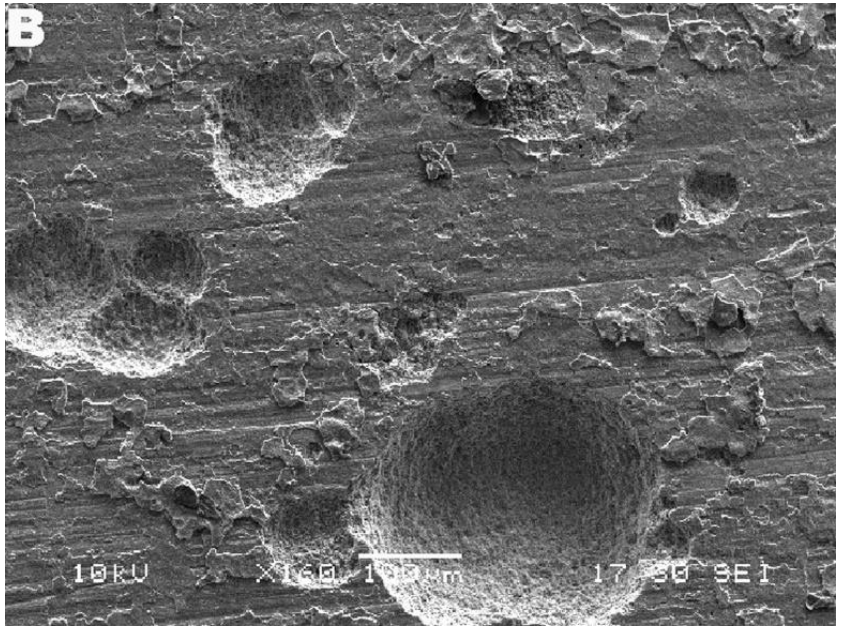
General facts about differential aeration corrosion

- Corrosion may be accelerated in apparently inaccessible places, because the oxygen deficient areas serve as anode, therefore, cracks serve as foci for corrosion.
- Corrosion is accelerated under accumulation of dirt sand, scale and other contamination. This is because accumulation of rust and scale restricts the access of oxygen and establish an anode to promote greater accumulation.
- Metal exposed to aqueous media under blocks of wood or piece of glass, which screen that portion of metal from oxygen.

3.b) Pitting corrosion

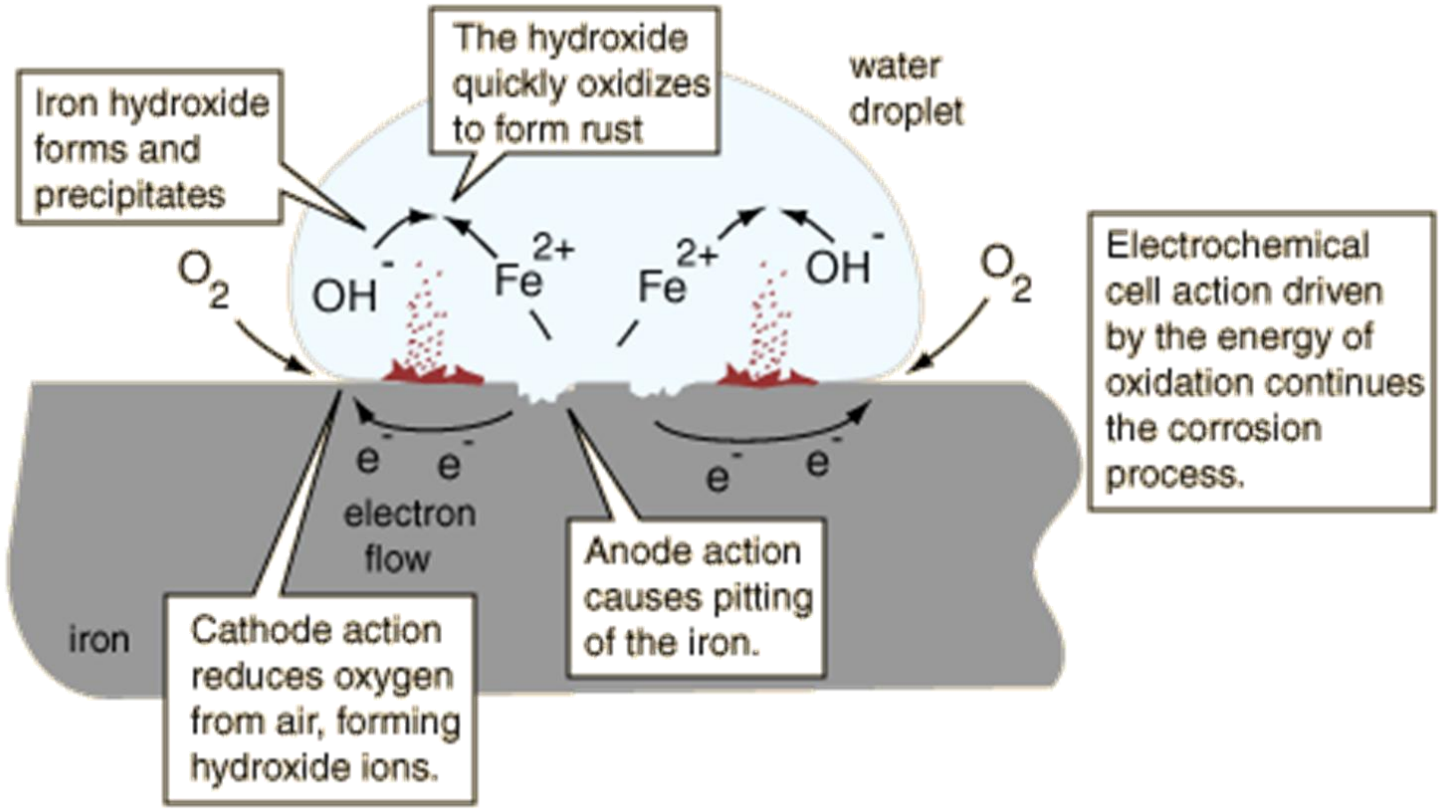
Pitting corrosion, is a form of extremely localized corrosion that leads to the creation of small holes in the metal.





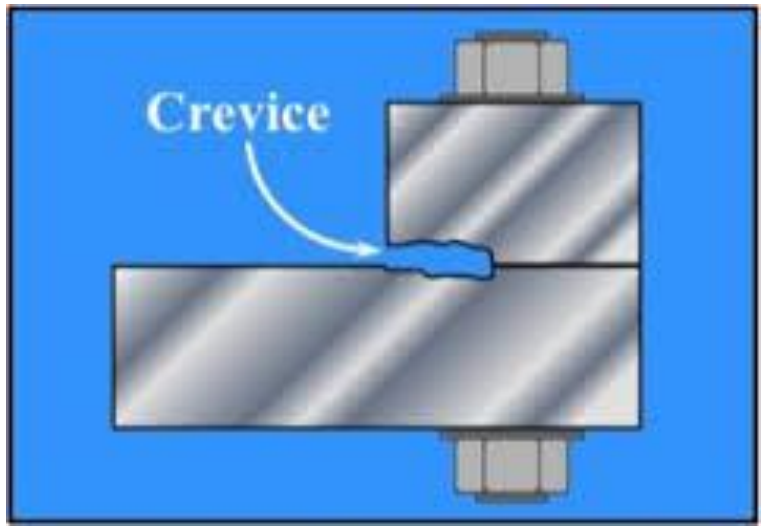
3.b) Pitting corrosion

Pitting Corrosion is an extremely localized corrosion mechanism that causes destructive pits.



3.c) Crevice corrosion

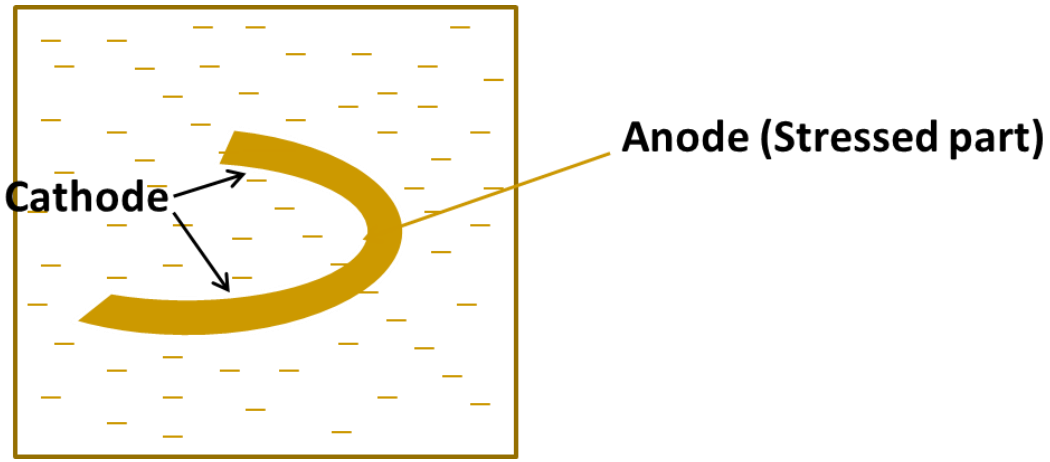
- Here, corrosion take place in crevices because solutions retained at this place and takes longer time to dry out.
- When this occurs, the attack is more severe at crevices.
- Crevices are formed because of the metal contact with another piece of the same or other metal or with a nonmetallic material.
- Corrosion in crevice is due to deficiency of O_2, acidity changes,.





d) Stress corrosion

Stress corrosion is combined effects of static tensile stress and the corrosive environments on a metal.



Dry or chemical corrosion:

- This occurs at dry conditions
- Corrosion is uniform
- It is a slow process
- It involves direct chemical attack
- Explained by adsorption mechanism

Wet or electrochemical corrosion:

- This occurs at wet conditions (electrolytic medium)
- Corrosion Is not uniform
- It is a rapid process
- It involves formation of electrochemical cells
- Explained by mechanism of electrochemical reactions

3. Cathodic protection

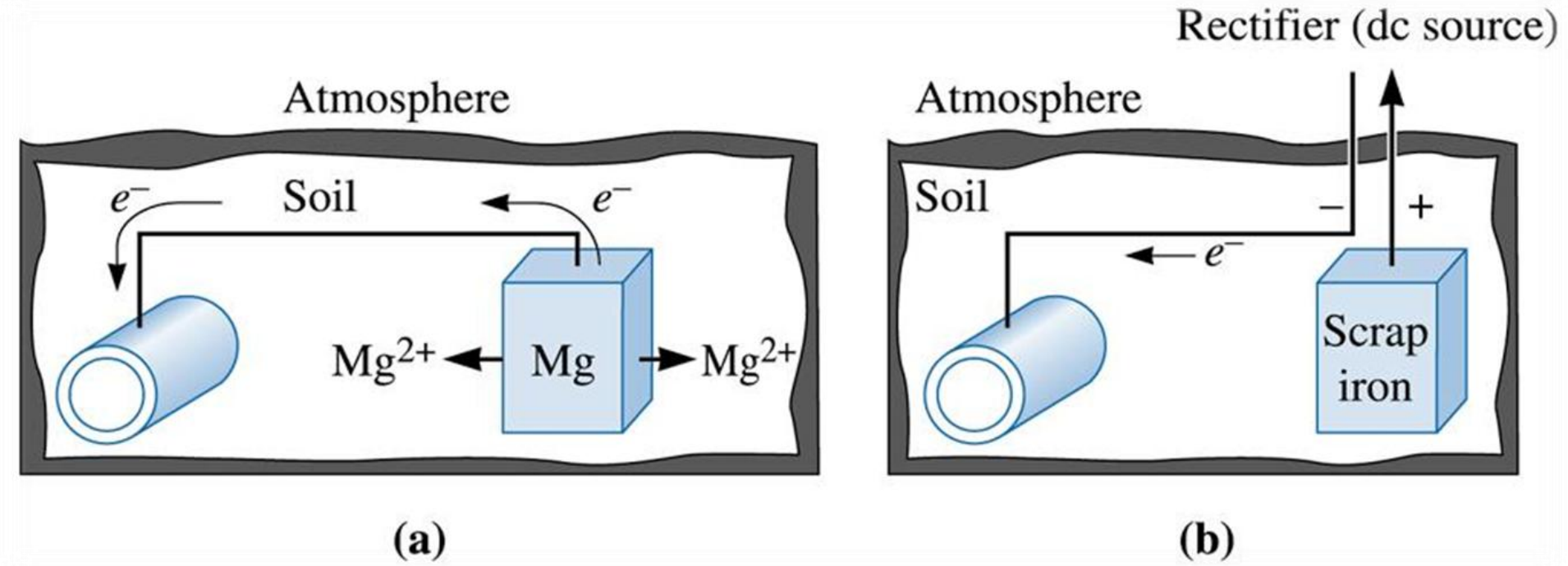
The method of protection given to a metal by forcibly making it to behave like a cathode

1. Sacrificial anodic protection:

In this method of protection , the metallic structure to be protected called “base metal” is connected to more anodic metal through a wire.

2. Impressed current cathodic protection:

In this method an impressed current is applied in the opposite direction to nullify the corrosion current, thus the anodic corroding metal becomes cathodic and protected from corrosion.



Cathodic protection of a buried steel pipeline:

(a) A sacrificial magnesium anode assures that the galvanic cell makes the pipeline the cathode.

(b) An impressed voltage between a scrap iron auxiliary anode and the pipeline assures that the pipeline is the cathode.

3. Cathodic protection

Examples: Zinc is attached to the steel hull of the vessel







Corrosion inhibitors

A substance which when added in small quantities to the aqueous corrosive environment, effectively decrease the corrosion of a metal.

Anodic inhibitors:

These inhibitors avoid the corrosion reaction occurring at the anode, by forming the sparingly soluble compounds. They are adsorbed on the metal surface , forming a protective film or barrier, thereby reducing the corrosion rate

Ex: Chromates , Phosphates , Tungstates

Corrosion inhibitors

Cathodic inhibitors:

In **acidic medium** corrosion may reduced by:

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

1. Slowing down the diffusion of H^+ ions by adding organic inhibitors like amines, mercaptans, heterocyclic nitrogen compounds.
2. By increasing the over voltage of hydrogen evolution by adding inhibitors like Antimony and Arsenic oxides

In **neutral medium** corrosion may reduced by :

$$\frac{1}{2}O_2(g) + H_2O(aq) + 2e^- \rightarrow 2OH^-$$

1. Eliminating oxygen from the corrosion medium by the addition of Na_2SO_3, Na_2S ;

Electroplating

Electroplating is a process of depositing a thin layer of a fine and superior metal (like Cr, Zn, Ni, Au etc.) over the article of a baser and cheaper metal (like iron), with the help of electric current.

Popular methods:

- Electroplating with Ni
- Electroplating with Cu

**Role of pH in electroplating



Electroplating – Steps:

- Before electroplating the metal surface is cleaned thoroughly. Firstly, an alkaline solution is used to remove grease and then it is treated with acid to remove any oxide layer. It is then washed with water.
- The article to be electroplated is made cathode since metallic ions are positive and thus get deposited on the cathode.
- The anode is made of pure metal, which is to be coated on the article.
- The electrolyte is the salt of the metal to be coated on the article.
- A direct (D.C.) current is passed through the electrolyte. The anode dissolves, depositing the metal ions from the solution on the article in the form of a metallic coating.

Electroplating with Ni:

- Electrolyte - Aqueous solution of nickel sulphate
- Cathode - Article to be electroplated (nail)
- Anode - Block of nickel metal.
- Dissociation of nickel sulphate: $NiSO_4 \rightarrow Ni^{2+} + SO_4^{2-}$

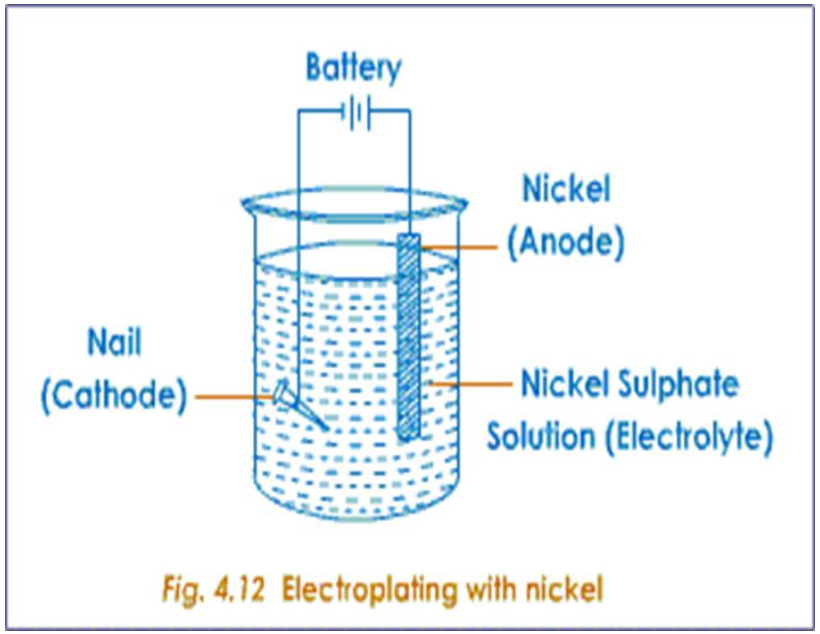


Fig. 4.12 Electroplating with nickel

Electroplating with Ni:

Reaction at Cathode (Reduction)	Reaction at Anode (Oxidation)
Ions at cathode: Ni^{2+}, H^{+}	**Ions at anode:** SO_4^{2-}, OH^{-}
Preferential discharge of Ni^{2+} ions takes place.	None of the above ions discharge at anode; instead Ni atoms of the anode lose electrons to form Ni^{2+} ions.
$Ni^{2+}(aq) + 2e^{-} \longrightarrow Ni^{0}(s)$	$Ni^{0}(s) - 2e^{-} \longrightarrow Ni^{2+}(aq)$
Nickel metal deposits on the article, i.e., nail.	Ni^{2+} ions are formed at anode.

END OF LECTURE

(a) Calculate the PB ratio for following oxides and predict if the layers are protective or not.

Metal Oxide	Metal Oxide Density ((g/cm^3))	Atomic weight of metal	Metal density (g/cm^3)	Molecular weight of metal oxide
ZrO_2	5.68	91.2	6.52	123.2
K_2O	2.13	39.09	0.862	94.18

(a)



Study the above two cases. In which case the corrosion rate is faster and why is it faster?

(b) Give reasons:

(i) Screw and nut made by same metal is preferred in practice to avoid corrosion.

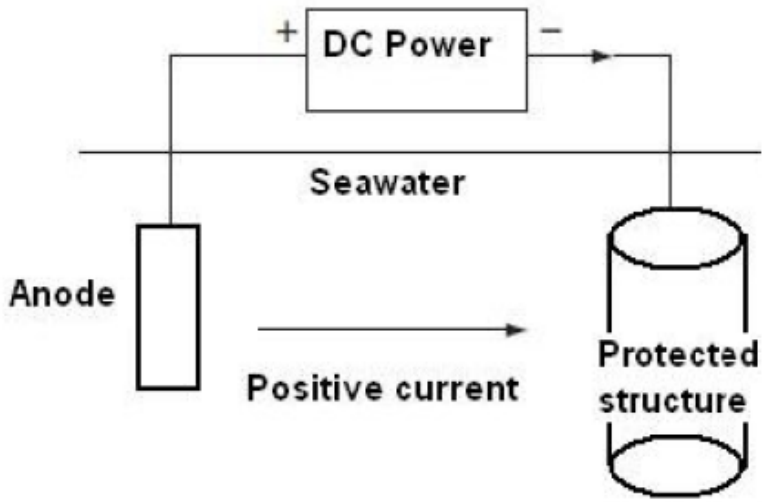
(ii) The part of metal sheet covered with a piece of wood undergoes corrosion as compared to the part free exposed to air.

Sol:

(i) To prevent galvanic (bi-metallic) corrosion

(ii) Differential aeration corrosion, less oxygenated part acts as anode. Corrosion occurs at anodic region.

(a) Identify the corrosion prevention process applied in following diagram. Write about the mechanism in the process.



Discuss the role of added sodium sulphite or hydrazine to boiler feedwater. Mention the reactions involved.

Ans:

Both are reducing agents and remove oxygen from water.

$2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$

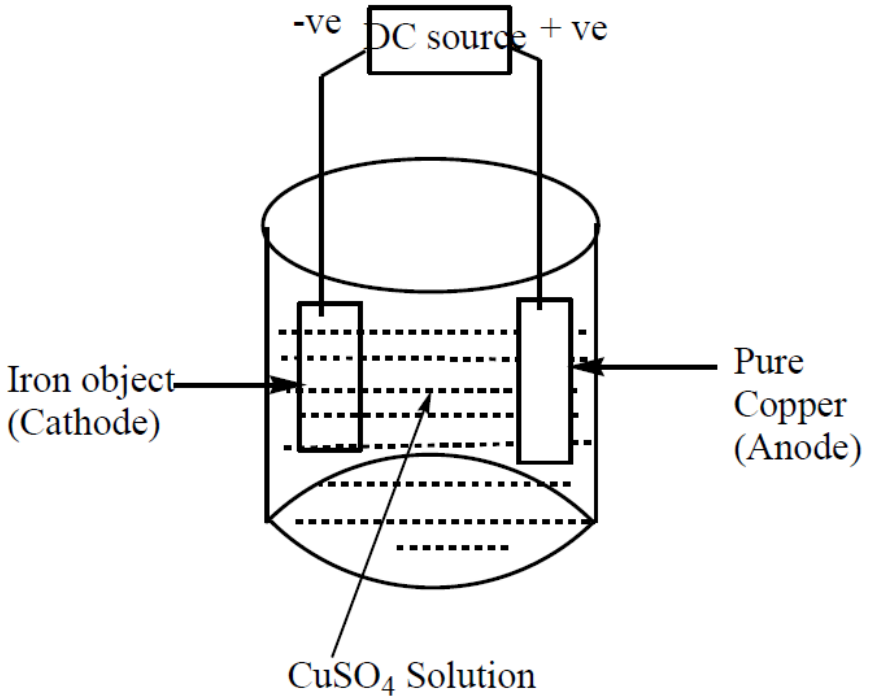
$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$

Calculate the pilling Bedworth ratio and find out which of the metal oxides films will be protective and why?

Metal	Metal Oxide	Molar volume of metal on surface (cm^3)	Metal oxide density (g/cc)	Metal density (g/cc)
Sn	SnO_2	16.26	6.95	7.30
Ni	NiO	6.58	6.67	8.91

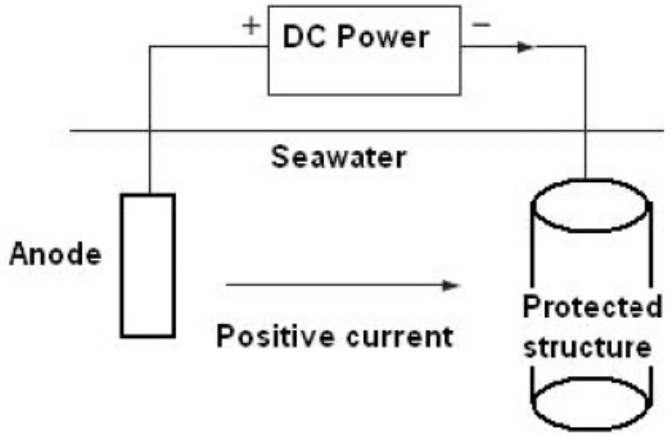
Draw a schematic diagram and explain in brief electroplating process of copper (Cu) over an iron (Fe) object.

Ans:



Cathode; Anode; Electrolyte

Identify the corrosion prevention process applied in following diagram. Write about the mechanism in the process.



Ans

Impressed current method

Cathodic protection

3.	Calculate PB ratio for the following metal oxides and predict the protection of formed coatings.	[3] (mark for formula, 3 marks for complete calculation)

Oxide	Oxide Density (g/cc)	A Wt. of Metal	Metal Density (g/cc)	Mol. Wt. of Metal Oxide
PbO	9.53	207.2	11.34	223.2
Li_2O	2.013	7	0.534	30

Sol: PB Ratio (PbO)= 1.28
PB Ratio(Li_2O)= 0.568

4.	Iron does not corrode when the coating is broken in a galvanized iron pipe but severe corrosion occurs if a coating of tin over iron is broken. Explain the fact with proper electrochemical reaction. **Ans:** Corrosion occurs at anode/ anodic region. Galvanization means coating of iron object by zinc. Zn is more anodic than iron, so in first case coating (in this case Zn) will act as anode and iron will act as cathode. Among tin and iron, tin is less reactive than iron. In second case coating (tin) will act as cathode and iron will act as anode. Case I **$Zn \rightarrow Zn^{2+} + 2e^-$ (anodic reaction, Oxidation)** $H_2O + 2e^- + \frac{1}{2} O_2 \rightarrow 2OH^-$ (Cathodic reaction, Reduction) Case II **$Fe \rightarrow Fe^{2+} + 2e^-$ (anodic reaction, Oxidation)** $H_2O + 2e^- + \frac{1}{2} O_2 \rightarrow 2OH^-$ (Cathodic reaction, Reduction)	[3] (explanation 1.5 marks, reactions 1.5 marks)

(b) Explain Caustic embrittlement in detail.

Sol: This is very dangerous form of stress corrosion, occurring in mild steel exposed to alkaline solution at high temperature and stresses. The failure is often associated with steam boiler and heat transfer equipment in which water of high alkalinity attacks the mild steel plates, particularly at the crevices near rivets.

$$Na_2CO_3 + H_2O \rightarrow 2\,NaOH + CO_2$$

Dilute alkaline boiler water flow into the minute hair-cracks and crevices by capillary action. Water evaporates and the concentration of caustic soda builds up. The concentrated alkali dissolve iron as sodium ferroate (Na_2FeO_2) in cracks and crevices

The sodium ferroate (Na_2FeO_2) decomposes, a short distances from its point of formation.

$$3Na_2FeO_2 + 4H_2O \rightarrow 6\,NaOH + Fe_3O_4 + H_2$$

$$6Na_2FeO_2 + 6H_2O + O_2 \rightarrow 12\,NaOH + 2Fe_3O_4$$

7.	What is the type of corrosion that may occur in following cases? (i) Riveted joints (ii) Iron tank filled half with water (iii) Steel pipe connected to copper plumbing (iv) Buries pipelines (v) Nail immersed in wood block (vi) Caustic embrittlement of steel	[3] Each carry 0.5 mark

(b) Identify the type of corrosion that may occur in following cases:

(i) Steel nut with aluminum bolt
(ii) Iron tank filled half with water
(iii) Steel pipe connected to copper plumbing
(iv) Nail immersed in wood block.

Sol: (i) Galvanic (ii) water-line (iii) Galvanic (iv) Differential aeration
(each carry 0.5 mark)

(c) Calculate PB ratio for the following metal oxides and comment on the protection of the formed coatings.

Oxide	Oxide Density (g/cc)	Atomic weight of Metal	Metal Density (g/cc)	Molecular Weight of Oxide
MgO	3.65	24.3	1.74	40.3
CaO	3.40	40.08	1.55	56.1
Al_2O_3	3.70	27.00	2.70	102.0

Sol: MgO PB ratio: 0.79 (non-protective)
CaO PB ratio: 0.3 (non-protective)
Al_2O_3 PB ratio: 1.37(Protective)
Each carry 1 mark

(a) What is dry corrosion? Discuss its various types.

Sol: Definition : Chemical attack by environment on metal (0.5 mark)
Types: (1) oxidation (ii) corrosion by other gases (iii) liquid metal corrosion
With some details in each category

(a) Study the below two cases. In which case the corrosion rate is faster and why is it faster?



Sol: (a) Corrosion rate is faster in (I) because of high electron density

(b) Discuss the wet corrosion mechanism in acidic environment with suitable diagram and reactions.

What is the type of corrosion that may occur in following cases?
(i) Riveted joints
(ii) Iron tank filled half with water
(iii) Steel pipe connected to copper plumbing
(iv) Buries pipelines
(v) Nail immersed in wood block
(vi) Caustic embrittlement of steel

Sol: (i) stress, (ii) waterline (iii) galvanic (iv) differential aeration (v) differential aeration (vi) stress